The complex Grothendieck constant

New bounds, machine-checked in Lean

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https://github.com/agalatan/grothendieck-lean

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$$1.401692002039 \;<\; K_G^{\mathbb{C}} \;<\; 1.404598989011$$

Both bounds are theorems of a Lean 4/Mathlib development, each resting on one explicitly stated family of externally certified numerical inputs (Section 3).

Abstract

Grothendieck's inequality bounds how much a semidefinite (vector) relaxation of a bilinear optimisation problem can exceed the true optimum. The best constant in the complex case, $K_G^{\mathbb{C}}$, has been known since 1987 to lie roughly in $[1.338, 1.405]$; its first two decimals were settled only in September 2026. We improve both bounds, to an interval of width 0.0029, and verify the arguments in Lean. This note explains the problem, the two constructions, exactly what the computer checks, the new inequalities and obstructions found along the way, and why the remaining gap is hard: the two sides are dual optimisation problems, and the obstruction on both is the nonlinear (cubic and higher) structure of bounded functions of Gaussian vectors.

1 The question

Let $A = (a_{ij})$ be an $m \times n$ complex matrix. Compare

$$C(A) = \sup_{|\varepsilon_i|=|\delta_j|=1} \Big|\sum_{i,j} a_{ij}\,\overline{\varepsilon_i}\,\delta_j\Big|, \qquad V(A) = \sup_{\|x_i\|=\|y_j\|=1} \Big|\sum_{i,j} a_{ij}\,\langle x_i, y_j\rangle\Big|,$$

the first over unimodular scalars, the second over unit vectors in any complex Hilbert space. Clearly $C(A) \le V(A)$. Grothendieck proved in 1953 that $V(A) \le K\,C(A)$ for a constant K independent of m, n and the dimension. The smallest such K is the *complex Grothendieck constant* $K_G^{\mathbb{C}}$.

Why it matters.

- *Optimisation.* $V(A)$ is a semidefinite program; $C(A)$ is a hard combinatorial problem. $K_G^{\mathbb{C}}$ is the best approximation ratio of the semidefinite relaxation, and rounding algorithms for it are exactly upper-bound proofs.
- *Functional analysis.* K_G is the norm of the identity between the injective and projective tensor norms on $\ell_\infty \otimes \ell_\infty$, and controls absolutely summing operators (Grothendieck's original setting).
- *Quantum information.* The real constant $K_G^{\mathbb{R}}$ is the largest ratio between quantum and classical values of two-player XOR games (Tsirelson); the finite-order constants $K_G(d)$ govern the nonlocality of qubit states.

History.

year	who	lower bound	upper bound
1953	Grothendieck		$K < \infty$
1984	Davie	1.33807	
1987	Haagerup		1.40490913… (Gaussian phase rounding)
Sep 2026	Guo, Fang, Lu	1.35584631827	
Sep 2026	Heilman, Jones, Malavolta	1.4	1.404898554746
Oct 2026	**this work**	**1.401692002039**	**1.404598989011**

Haagerup's integral $1/\gamma^\dagger = 1.404575934\ldots$, $\gamma^\dagger = \int_0^{\pi/2} \cos^2 t\,(1+\sin^2 t)^{-1/2} dt$, has been proposed as the exact value. It lies inside our interval, $2.31 \cdot 10^{-5}$ below the upper bound. For the real constant the corresponding recent results are $1.773 \le K_G^{\mathbb{R}} \le 1.7799$ (Heilman–Jones–Malavolta, October 2026).

2 Two dual constructions

Both bounds use the same Gaussian picture. Let $Z = (z_1, \ldots, z_n)$ be a standard complex Gaussian vector and decompose $L^2(\mathbb{C}^n, \gamma)$ into the *Wick chaoses* $H_{p,q}$ (polynomials of bidegree (p, q) in $z, \bar{z}$, orthogonalised). Under the rotation $z \mapsto e^{i\theta} z$ every relevant object has *charge one*, so only the bidegrees $(1,0), (2,1), (3,2), \ldots$ (degrees $1, 3, 5, \ldots$) appear.

2.1 Upper bounds: rounding schemes

Given unit vectors, project a Gaussian onto them and round: $\varepsilon_i = f(\langle G, x_i\rangle)$ with $|f| \le 1$. The correlation of two rounded values depends only on $t = \langle x_i, y_j\rangle$,

$$\mathbb{E}\big[\overline{f(\cdot)}\, g(\cdot)\big] = H(t) = \sum_{k \ge 0} c_k\, t|t|^{2k}.$$

If $H(t) = c\,t + r(t)$ with $\|r\|_A = \sum_k |c_k - c\,\delta_{k0}| < c$, a tensor-power argument (Krivine's trick) gives $K_G^{\mathbb{C}} \le 1/(c - \|r\|_A)$. Haagerup's choice $f = \operatorname{phase}$ gives 1.40491. Our rounding is a smoothed two-coordinate phase rounding, $F_0 = \operatorname{phase}(z + \sqrt{1+a^2}\, v(|z|^2)\, w)$ with a twelve-term Laguerre profile v, multiplied by the cubic Taylor polynomial of a damped polynomial phase φ: $T = F_0 P_3(\varphi)$, $P_3(x) = 1 + ix - x^2/2 - ix^3/6$. The identity $|P_3(x)|^2 = 1 - x^4(3 - x^2)/36$ and $|\varphi| < 1$ give $|T| \le 1$, and a disk-valued rounding is as good as a unimodular one. Its kernel is $H(t) = \sum E[n,m]\, p(t)^n (-\tau(t))^m$ with an odd polynomial p of degree 101 and $\tau(t) = (t^3 + a^2 t^7)/(1 + a^2)$. Compared with $c\,t$, $c = 625000/877873$, the residual is at most $\delta = 554771095509/(5 \cdot 10^{17}) \approx 1.10954 \cdot 10^{-6}$:

$$K_G^{\mathbb{C}} \le \frac{1}{c - \delta} = \frac{43893650000000000000000}{31249951298143407222764 3} < 1.404598989011.$$

The whole difficulty is to keep *every* coefficient c_k, $k \ge 1$, small at once: improving the cubic coefficient c_1 tends to spoil $c_2, c_3, \ldots$, and the infinite tail has to be bounded too.

2.2 Lower bounds: Hermite projection games

A lower bound needs an operator T whose *vector* value is close to 1 while its *scalar* value is small. We use

$$T = P_{1,0} - \epsilon\, P_{2,1}, \qquad \epsilon = \tfrac{149}{200},$$

the projection onto linear functions minus a multiple of the projection onto the cubic chaos.

- *Vector value.* With $u_\ell = z_\ell/\|z\|$, $\sum_\ell \langle u_\ell, Tu_\ell\rangle \geq 1 - (1+\epsilon)/(n+1) \to 1$.
- *Scalar value.* For all $|f|, |g| \leq 1$, $|\langle f, Tg\rangle| \leq B = 0.71342349$.

A finite-partition transfer turns this into matrices, so $K_G^{\mathbb{C}} \geq 1/B > 1.401692002039$.

The scalar cap is the hard part. Writing $h = \frac{1}{2}(f+g)$, $k = \frac{1}{2}(f-g)$,

$$\Re\langle f, Tg\rangle = t^2 - \epsilon\|P_{2,1}h\|^2 - \|P_{1,0}k\|^2 + \epsilon\|P_{2,1}k\|^2, \qquad t = \|P_{1,0}h\|.$$

Only the last term is positive: we must show that a *bounded* function cannot have too large a cubic part without paying elsewhere. After a rotation $P_{1,0}h = t\, z_1$, and the cubic part of k splits into pieces that involve z_1 (*visible* sectors) and a *remote* residual $q \in H_{2,1}$ in the other coordinates. The proof then has three layers.

1. **Row reduction.** For each of 19 profiles r, the contribution of the visible sectors is the supremum of an explicit finite-dimensional Gaussian integral, a convex one-variable *potential* Ψ_r, evaluated at the scale of the remote residual. One gate variable is shared across rows.
2. **The law of the remote cubic.** What remains is a statement about the distribution of $U = |q|^2/\mathbb{E}|q|^2$ for an arbitrary cubic Wick polynomial in arbitrarily many variables. We use exact identities and inequalities valid for *every* such cubic: the fourth-moment cap $\mathbb{E}U^2 \leq 62$; the sixth-moment bound $\mathbb{E}U^3 \leq 9m - 12 + 40.45313\,(m-2)^{3/2}$, $m = \mathbb{E}U^2$, which rests on thirteen tensor-contraction facts including the Wick expansion of the sixth cumulant; a first-moment bound; and families of *Stein cuts* $\mathbb{E}\,G(U) \geq 0$, obtained by Gaussian integration by parts and completing squares against spectral budgets, a covariance order and *contraction-Gram budgets* (positive semidefinite Gram matrices of the coefficient tensor).
3. **A finite certificate.** The parameters are covered by 7,939 boxes in (t, τ, x); on each box a positive combination of rows and node certificates (336 joint nodes, 9,237 row nodes, each a polynomial majorant checked in exact rational arithmetic) bounds the scalar value by B.

2.3 The two problems are one

For the real constant, Heilman, Jones and Malavolta proved an exact minimax: over rounding schemes μ with kernel coefficients $c_m(\mu)$ and Hermite projection games $Q = \sum_m q_m \Pi_m$,

$$\frac{1}{K_G} = \sup_\mu \Big[c_1(\mu) - \sum_{m\geq 3} |c_m(\mu)|\Big] = \inf_Q \ \mathrm{OPT}(Q),$$

and the same duality holds in the complex case on charge-one bidegrees. An upper bound is a good point of the convex set of achievable kernels; a lower bound is a supporting hyperplane of that set. There is *no intrinsic gap*: the gap between our bounds is entirely the gap between restricted constructions and the cost of certifying them.

3 What exactly the computer checks

Proved in Lean	**Assumed (certified outside Lean)**
Lower bound. `lower_bound_of_nodeCaps : NodeCaps → IsGrothendieckBound K → 1.401692002039 < K` the witness T on L^2 of the Gaussian measure and its vector value; the transfer to matrices; the polarisation identity; the per-row Gaussian reduction of $\langle f, Tg\rangle$ for every unimodular pair (frames, fibre decomposition, Parseval, gates); all moment, Stein, spectral and contraction-Gram inequalities for every circular cubic, including the sixth-cumulant Wick identity; the finite certificate (one `native_decide`)	`NodeCaps`: the 19 potentials Ψ_r (defined in Lean) lie below their stored rational caps at the tabulated nodes. These caps come from exact-arithmetic certificates (interval integration and anchored quadratic majorants), replayed independently but not yet formalised.
Upper bound. `fresh_upper : DirectHyps E EX → IsGrothendieckBound (4389365000000000000000/3124995129814340722276 43)` the transfer inequality (tensor powers); kernel composition and locality; the head of the kernel through degree 501 from 21,168 stored energies by a verified interval evaluator (one `native_decide`); the tail bound; the endpoint	`DirectHyps`: the tables E, E_X are the energies of the rounding $T = F_0P_3(\varphi)$ and of its truncation X (validity, summability, parity); the comparison $\sum \lvert E - E_X\rvert \le 1065637497/10^{15}$ (96.04% of the residual budget); a weighted square-sum bound $7/4$ at radius $4149/4000$, giving the tail $36748/10^{12}$ (3.31%); the table error $6847009/(5 \cdot 10^{17})$ (0.001%). The head computed in Lean, $7143/10^{12}$, is 0.64%.
Unconditional upper bound. `haagerup_bound : IsGrothendieckBound (140578/100000)` Haagerup's identity for the phase rounding, $\mathbb{E}[\overline{\mathrm{phase}\langle g,x\rangle}\ \mathrm{phase}\langle g,y\rangle] = \sum_k \frac{\pi}{4}\frac{(1/2)_k^2}{k!(k+1)!} z\lvert z\rvert^{2k}$ (Euler integral, Gaussian integrals, Beta integrals, reduction to two complex Gaussians); the transfer through the cubic preprocessor $p(t) = \frac{2267}{2500}t - 0.09319996\,t^3$; 4000 exact terms (one `native_decide`) and a telescoping tail	nothing: this bound has no hypothesis. It is weaker than the certified bound above, and than Haagerup's 1.40491, because only a cubic preprocessor and a finite head are used.

Axioms: `propext`, `Classical.choice`, `Quot.sound`, and exactly three pinned `native_decide` facts. No `sorry`; CI builds everything and checks the axiom report (162 theorems). The development has about 28,000 lines of hand-written Lean and 11,000 lines of generated certificate data. The lower bound is close to fully formal: only one-variable numerical caps remain. The sharp upper bound still rests mainly on an external analytic comparison lemma; a weaker upper bound, 1.40578, is fully formal.

4 Discoveries along the way

Status labels: **[Lean]** machine-checked in this repository; **[P]** written proof; **[P,C]** proof with an exact or interval-arithmetic certificate replayed independently; **[F]** floating-point evidence only.

4.1 Inequalities for cubic Gaussian chaos

Throughout, q is a circular cubic Wick polynomial (bidegree $(2,1)$) in any number of complex Gaussian variables, normalised by $\mathbb{E}|q|^2 = 1$, and $U = |q|^2$.

- **Sharp fourth moment [Lean].** Every real third-chaos variable satisfies $\mathbb{E}F^4 \le 93\,(\mathbb{E}F^2)^2$, with equality for $x^3 - 3x$. Hence $\mathbb{E}U^2 \le 62$.
- **Sixth moment [Lean].** $\mathbb{E}U^3 \le 9m - 12 + C_6(m-2)^{3/2}$ with $m = \mathbb{E}U^2$ and $C_6 = 40.45313$. The Wick monomial mixed with Gaussian noise attains 40.295, so the constant is sharp to 0.4%. The inequality is *false* without circularity: two real cubics $He_3(x_0)/\sqrt{12}$, $He_3(x_1)/\sqrt{12}$ give $m = 47$ and $\mathbb{E}U^3 = 13506 > 12622.55$.
- **The sixth cumulant as a sum over graphs [Lean].** The Wick identity for κ_6 of a cubic chaos is proved by expanding $\mathbb{E}F^2, \mathbb{E}F^4, \mathbb{E}F^6$ over closed cubic multigraphs, one Gaussian integration by parts per edge ($69 + 15 + 3$ certified steps). The disconnected graphs cancel, and the six connected cubic multigraphs on six vertices give the tensor contractions.
- **Contraction-Gram budgets [Lean].** A new family of universal inequalities between the chaos components of U and of q^2. With $X = \Gamma(q,\bar{q})/3 - 1$, $c = \Gamma(q,q)/3$, $r = \Re(c\bar{q}^2)/U$, $P = \mathbb{E}[XU]$, $Q = \mathbb{E}[rU]$, for example

  $$\mathbb{E}X^2 + \tfrac{1}{10}\mathbb{E}r^2 \le \tfrac{34}{45}P - \tfrac{1}{18}Q.$$

  Each budget is certified by two positive semidefinite 2×2 Gram matrices of contractions of the coefficient tensor.
- **First moment [P,C].** $\mathbb{E}|q| \le 0.97073$, from a two-plane small-ball estimate (a winding argument on S^3 and a horizontal Lipschitz bound). The sharp value $\sqrt{\pi}/2 = 0.8862$, which the Gaussian attains, is open.

A common budget for the same tensor. The sixth moment and the Stein fluctuations cannot be optimized independently: both depend on the same contraction invariants. We now formalize this connection in Lean, including the real/complex normalization and a finite checker that reserves part of the contraction budget before accepting a Stein cut. Signed reserve coefficients are retained. The analytic theorem is proved; the stronger external joint-moment certificate still needs its complete node and cover integration. Retaining the upper bound on the negative cumulant contraction yields a stronger parameterized moment family. An exact transport identity then upgrades a node's majorant when its budget permits a larger moment-cut weight, without a new half-line search. These two refinements are also formalized in Lean.

4.2 The remote functional

Let $L = z_1(a + b|z_1|^2)$ with the constants of the witness, and for a cubic G put $\Lambda(G) = \mathbb{E}|L+G| + \mathbb{E}|L-G| - \mathbb{E}|G|^2/\epsilon$ and $S = \sup_v\{2\mathbb{E}|L + \sqrt{v}U| - v/\epsilon\}$, U a complex Gaussian. The conjecture $\Lambda(G) \le S$ for all cubics ("Conjecture K") is the analytic heart of the lower bound.

- **A one-function reformulation [P].** Circular symmetry and $|a+b|+|a-b| = 2\max(|a|,|b|)$ give exactly

  $$\Lambda(G) = 2\mathbb{E}|L| + \pi\,\mathbb{E}\big(|\Re G| - |\Re L|\big)_+ - \mathbb{E}|G|^2/\epsilon,$$

  so $\sup_G \Lambda - 2\mathbb{E}|L| = \sup_t\big[\frac{\pi^2\epsilon}{4}\|P_{2,1}t\|^2 - \pi\mathbb{E}(|t|\,|\Re L|)\big]$ over real functions $|t| \le 1$. All of the gain sits near the hyperplane $\Re z_1 = 0$.
- **The Gaussian is a strict local maximum [P,C].** At the Gaussian point every direction that involves z_1 has negative second variation, certified in interval arithmetic for seven sector coefficients. The Hessian's kernel is the entire z_1-independent cubic sector; a negative fourth-order term controls those genuine cubic directions.
- **The independent sector [P,C].** For every cubic q not involving z_1, $\Lambda(q) < S + 3.6 \cdot 10^{-7}$. Moment inequalities alone stall about 10^{-5} above S, because $h(x) = 2\mathbb{E}|L + \sqrt{x}|$ is a

Bernstein function and the three-moment relaxation is maximised by a two-point law [P]. Going below that floor needs the Stein and spectral cuts.

- **Two natural shortcuts are false [P,C].** $|q|^2$ does not dominate $\mathbb{E}|q|^2 \cdot \mathrm{Exp}(1)$ in convex order, and there is no Gaussian Laplace domination: $Q = X(4(|X|^2-2)+3i(|Y|^2-1))/\sqrt{41}$ has $\mathbb{E}e^{-20|Q|^2} < 0.0453 < 1/21$.

4.3 Actual responses

Instead of bounding Λ for all cubics, one can use that the cubic comes from a bounded response. Write F for the cubic part of the response and $\mathcal{N}$ for the value of the witness on it.

- **Exact support [P].** The paired support $\frac{1}{2}(\mathbb{E}|A+B| + \mathbb{E}|A-B|)$ is exactly the pointwise support function of the complex response set $\{(h,k) = (e^{i\alpha}\cos\theta,\ ie^{i\alpha}\sin\theta)\}$, because $(|A+B|+|A-B|)^2 = 4\max_\theta |A\cos\theta - iB\sin\theta|^2$.
- **Concentrated cubics are harmless [P,C].** At $\epsilon = 1891/2500$ on the central rows $0.84 \le t \le 0.851$, $\mathcal{N} < 0.71325 < 1/1.402$ whenever the cubic part of the response concentrates enough energy on a structured subspace:
  - 80% on one diagonal Wick direction;
  - 90% on one mixed monomial $X^2\bar{Y}$, or on a whole slice $\{X^2\bar{A}\}$;
  - 93.5%, 96.2%, 97.6%, 99.2% on any cubic in $2, 3, 4, 6$ remote coordinates;
  - 95.5% to 99.95% on $H_{2,0}(A)H_{0,1}(B)$ with $\dim A \le 24$, via a sharp support theorem whose extremiser has law $R_m G/\sqrt{m(m+1)}$, $R_m \sim \Gamma(m,1)$.

  The remainder of the cubic may be arbitrary and fully coupled.
- **The obstruction to covering [P].** $Q_N = X_1(\sum_{j\le N}|X_j|^2 - N - 1)/\sqrt{N+1}$ keeps an influential direction while its projection onto every fixed-dimensional core tends to zero. So "small core or Gaussian" is not an exhaustive dichotomy.
- **How far from the benchmark structured cubics sit [F].** Remote-only payoffs relative to S:

sum of N slices, $N =$	1	2	4	8	16	24	∞ (Gaussian)
$S - \sup\Lambda$ (10^{-3})	1.68	1.42	1.08	0.73	0.43	0.32	0

  The tested concentrated cubics leave room of order 10^{-3}; these diffuse sums approach the benchmark. This evidence does not classify all near-maximizers.

4.4 Limits of the witness method

- **Ceilings [F] and [P,C].** The norm of the cubic witness $P_{1,0} - \epsilon P_{2,1}$ is about 0.71314 at its best $\epsilon \approx 0.756$ [F]: no certificate for it exceeds 1.40226. Explicit "chirp" responses prove that games of degree ≤ 5 cannot exceed 1.4036705, and of degree ≤ 7 cannot exceed 1.4044895 [P,C].
- **Extending the pattern fails [P,C].** The full alternating game $\sum_{k\ge 0}(-1)^k P_{k+1,k}$ has $\sup_n \|\cdot\|_{\infty\to 1} = 1$, attained by explicit unimodular responses (one in complex dimension 308,835 is replayed). So better high-degree games must be re-optimised against new responses, not extrapolated.
- **No duality gap, but no free lunch [P].** By the minimax of Section 2.3, combining a pool of actual responses truncated at level D, value v_D, with a witness certified to $v_D + \delta$ and a

mixture whose omitted coefficient tail is at most τ, gives $1/(v_D + \delta) \le K_G^{\mathbb{C}} \le 1/(v_D - \tau)$. The interval for $1/K_G^{\mathbb{C}}$ has width exactly $\delta + \tau$.

5 Why the gap is hard

The witness class caps the lower bound. A game supported on degrees $\{1, 3\}$ sees only the cubic constraint on roundings. Even a perfect evaluation of the norm of $P_{1,0} - \epsilon P_{2,1}$ cannot certify more than about 1.40226 (numerical, best $\epsilon \approx 0.756$). Proved obstructions are known for higher degrees: every game of degree ≤ 5 is capped at 1.4036705, and every game of degree ≤ 7 at 1.4044895, both below Haagerup's value. Reaching the upper end of the interval therefore needs games of degree ≥ 9; in the real case the certified game uses all odd degrees up to 21.

Certifying higher degree is the real bottleneck. Two different certificate architectures, gated rows and support functions, both lose about $2 \cdot 10^{-2}$ on the best degree-5 game. The reason is structural: a bounded function with a large cubic part is forced to have a degree-five part too. Through the visible coordinate, the cubic part sees the average of the remote coefficient profile and the quintic part its variation, and boundedness makes the profile vary. A certificate that treats the chaos degrees as independent cannot see this coupling. The coupling is *driver sharing*: in the dimension-limit model the best degree-5 game has norm 0.712464 ($1/N = 1.40358$) when its degree-5 mixed sectors $(|z_1|^2 - 1)W$ and $z_1^2\overline{W}$ are driven by the same remote cubic W as the cubic sector, and 0.724753 when they are driven by independent copies; the optimum then switches the degree-5 part off. So a higher-degree certificate must let one remote residual feed several degrees through the visible coordinate. The cubic rows already do this for their degree-3 sectors; extending them to the degree-5 companions of the same residual is the next construction [F].

Why the complex case is harder than the real one. For real ± 1 responses, $h = \frac{1}{2}(f + g)$ and $k = \frac{1}{2}(f - g)$ have disjoint supports, so the two sides of a game decouple at every degree. For complex unimodular responses $|h|^2 + |k|^2 = 1$ and $\Re(h\bar{k}) = 0$ force $f = e^{i\alpha}e^{i\theta}$, $g = e^{i\alpha}e^{-i\theta}$: one shared phase field α drives both sides, and they never separate. The same shared phase is why marginal laws do not determine the relevant nonlinear correlations: $(H(X), X)$ and $(H(Y), X)$, with $H(X) = X(|X|^2 - 2)/\sqrt{2}$ and X, Y independent, have identical marginals and covariances, but $\mathbb{E}[\bar{r}\, q|q|^2]$ equals $8\sqrt{2}$ in the first case and 0 in the second.

6 Open problems

1. Formalise the node caps (`NodeCaps`), which would make the lower bound fully machine-checked.
2. Formalise the rounding behind the upper bound: its Hermite expansion, the comparison lemma and the contour tail bound.
3. Certify a game of degree ≥ 5 with a *multi-grade fibre*: one bounded function of the visible coordinate and one remote direction per degree, so that the cross-degree coupling is kept. This is the route to a lower bound above 1.403.
4. Run a complex primal/dual search, alternating linear programs over games of degree $\le D$ and pools of roundings, to locate $K_G^{\mathbb{C}}$ to about 10^{-4} and to test whether Haagerup's value $1/\gamma^{\dagger}$ is the answer.
5. Prove the sharp first-moment bound $\mathbb{E}|q| \le \frac{\sqrt{\pi}}{2}\|q\|_2$ for cubic Wick polynomials. It is natural, it appears to be open, and it is the largest single lever on the law side.

How this was made

The constructions and certificates were found by several AI research systems working on a shared manuscript, cross-checking each other's packages by independent replay. The endpoint theorems have the explicit Lean hypothesis lists in Section 3, and their axioms are audited in CI. The discoveries labelled [P] and [C] also include paper proofs and external exact or interval certificates; they are not all formalized in Lean. The research manuscript and the certificate packages are kept outside the repository.

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